Virtual University of Pakistan

Bio302 Molecular Biology

Final term Past Papers

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Bio302

Today 302 paper two use of nitrogen mustered gas.

Kind of bond with the potential for hyrolytic cleavage DNA.

What are psi and phi values in a helix structure and mention value.

How many classes of splicing found in cell.

Transition to open complex. What are the two RNA polymerase present in the eukaryotic othr than RNA pol II and how the different from RNa pol II

Questions:
Nuclear Scaffold, bacterial core Enzyme

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What will happen to cellular machinery and function if pyrimidine dimer is not removed? Some MCQs:

CTD have series of repeats…….?

Pyrimidine dimers are formed by……. UV light, Infra-red??? Suicide Enzyme…..????

Miss pair nucleotides per 100000 or 10^5 MuTS

Homologs

5 Splice site recognize by U1 snRNP and U6 or opposite ???

Inchworming???

Ecoli predominant Sigma factor

Cells are preffered to grown in N14, Pb, Cs, Rs?? ssDNA is needed for RNA helicase, DNA helicase, RNA helicase winding, RNA helicase unwinding???

Indel Mutation is caused by (reagent) ????

Mismatch repair system ????

Mutation ki type

Splicing ki type

Tmplete prinmer junction

Secondary structure of RNA

termination sequence Function

of RNA ploymdrase

How RNA ploymerase 2 differ frm 3,1 Transition

to open complex in transcription

spl!cosome

Major nd m!nor groove

D!eastr bond

!ntron splicing I and rna prcesng evnt

Differentiate endo and exo nucleases causing agent

Ctd ta!l 117

Elongated pokymerase !n 118

H!gh regulated level of transcrptn !n 116

Torp!do model !n 119

two use of nitrogen mustered gas.

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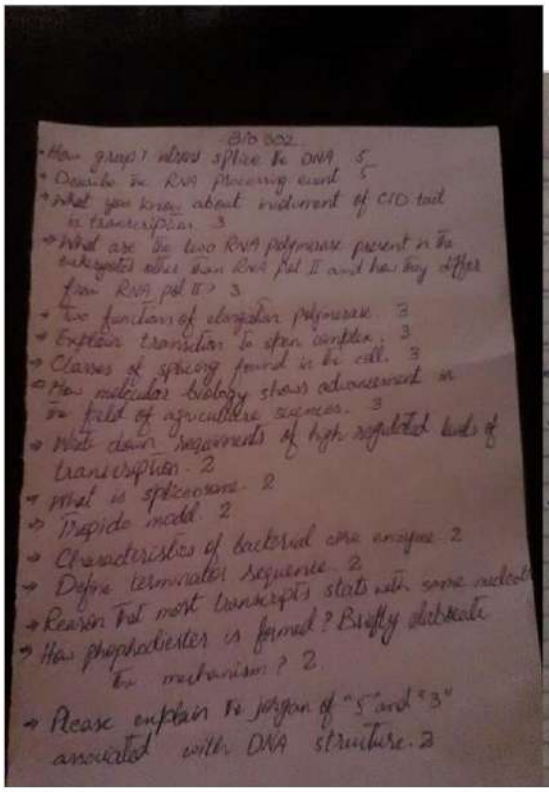
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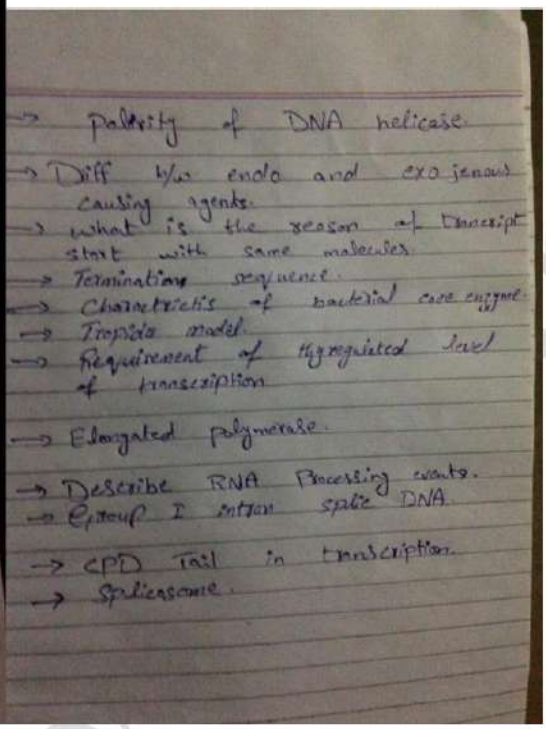
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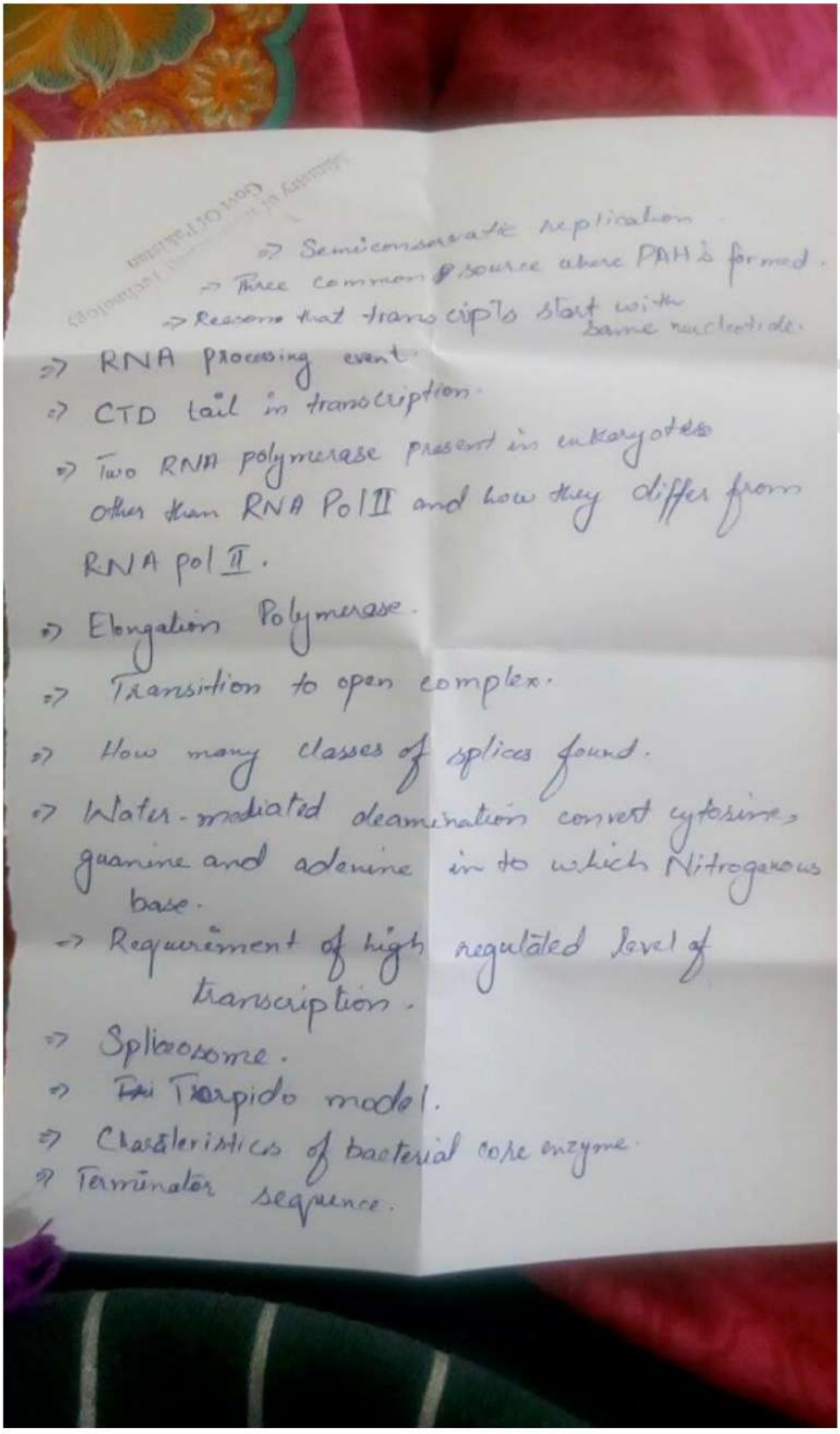
8/0 002

- How group I introns splice the DNA. 5
- Describe the RNA processing event 5
- What you know about inhibitors of CTD tail in transcription. 3
- What are the two RNA polymerase present in the eukaryotes other than RNA pol II and how they differ from RNA pol II? 3
- Two functions of elongation polymerase. 3
- Explain transition to open complex. 3
- Classes of splicing found in the cell. 3
- How molecular biology shows advancement in the field of agriculture sciences. 3
- With clear sequences of high regulated levels of transcription. 2
- What is spliceosome. 2
- Trapide model. 2
- Characteristics of bacterial core enzyme. 2
- Define terminator sequence. 2
- Reason that most transcripts start with same nucleotide
- How phosphodiester is formed? Briefly elaborate the mechanism? 2.
- Please explain the jargon of "5' and "3" associated with DNA structure. 2



→ Polarity of DNA helicase.
→ Diff b/w endo and exo genous causing agents.
→ What is the reason of transcript start with same molecules.
→ Termination sequence.
→ Characteristics of bacterial core enzyme.
→ Trapida model.
→ Requirement of upregulated level of transcription.
→ Elongated polymerase.
→ Describe RNA Processing events.
→ Group I intron splic DNA.
→ CPD Tail in transcription.
→ Spliceosome.

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=> Semiconservative replication.
-> Three common source where PAH is formed.
-> Reasons that transcripts start with same nucleotide.
=> RNA processing event.
=> CTD tail in transcription.
=> Two RNA polymerase present in eukaryotes other than RNA Pol II and how they differ from RNA pol II.
=> Elongation Polymerase.
=> Transition to open complex.
=> How many classes of splices found.
=> Water-mediated deamination convert cytosine, guanine and adenine in to which Nitrogenous base.
-> Requirement of high regulated level of transcription.
=> Spliceosome.
=> Tornpido model.
=> Characteristics of bacterial core enzyme.
=> Terminator sequence.

CRISPR stands for? 2 marks Terpedo
model. 2 marks
mRNA has short life span write experiment. 3 marks
Operon model? 5 marks Group 1
intron splicing. 3 marks
Splicing. 5 marks.

Gene expression controlled by regulatory proteins. 10 marks
RNA editing explain. 10 marks. mRNA short life experiment
intron splicing 1 CRIspr stand For enlist th name of splicing
method

.terminator sequence
Splisosome pathway..differnt pathway
CRIsPR. .application

splicing 10 gene
expression 10 crisp stand
for 2 mRNA with
experiment 5 tredo model

torpedo model? CRISPR stands for? mRNA
has short life span write experiment 1 short
yaad nhi aara.. :/ :P long 5 marks:
operon model?
group 1 intron splicing.
long 10 marks splicing gene expression controlled by
regulatory proteins. mcqs tough thy.. mostly enzymes
aurstrands k baary mai thy. short qustn: CRISPR stand for?
(2) terminating squence? (2) mRNA span short time (3) RNA
processing event? (5) opendro model? (5) long qustn:
gene of expression (10)
RNA editing. (10)
Write down the requirments of high regulated level of transcription?
CRISPRs stand for?
mRNA has short life span? spliceosme?
Describe RNA processing event?

:Explain Operon model?
RNA editinf explain?
What is CRISPRs? work? application?
Crispr
Crisper+ it's work +applications (10marks)
RNA elongation(10marks)

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Operon model
Splicesome
Write down the requirments of high regulated level of transcription?
CRISPRs stand for?
mRNA has short life span? spliceosme?
Describe RNA processing event?
Explain Operon model?
RNA editinf explain?
What is CRISPRs? work? Application

What is CRSPRs
What is spliceosomes
What is toredo model
Enlist slide pathways and describe It individually
Describe RNA process events
Describe operon model in 5 points
What is csrps??how does it works?? What r its implications and aplications

CRISPR stands? 2
Torpedo model 2
RNA has short life span. Explain experiment. 5
What are spliceosome? 3
Operon model. 5
How transition to open complex work? 3
Note on CRISPR? 10
SPLICING PATHWAY
A302. 18/8/2017 CRISPRs
.
Name two proteins found in E.coli.2 mrKs
Gene regulation. 10mrks
Lac opern. 5marks
Initiation complex..3mrks
RNA editing 10mrks.
Mcqs bhooot ajeeeb bs 2 3 mcqs past sy ay. R

Bio 302

Most important questions for Bio 302
Q.CRISPR stands for? 2 marks

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Q.Terpedo model. 2 marks

Q.mRNA has short life span write experiment. 3 marks
Q. Operon model? 5 marks
Q. Group 1 intron splicing. 3 marks
Q. Splicing. 5 marks
Q. Gene expression controlled by regulatory proteins. 10 marks
Q: RNA editing explain. 10 marks

Q.terminator sequence 2 marks
Q.RNA processing event? (5)
Q.What is spliceosme? 2 marks

Q: What is CRISPRs? work? application? 10 marks

Q. : Write down the requirments of high regulated level of transcription? 5 marks

Q. RNA elongation (10marks

Bio 302

Bio 302,,,,5pm

write two protein name in E.coli use for lactose.(2)
CRISPR stands for?(2) write name of splicing
classes.(2) what factors required for transcription in
eukaryotes?(3) how RNA has short life span?write
experiment(3)
write five points of operon model.(5) write
RNA processing events.(5)
write note on splicing pathway(10) how gene expression effect on regulatory protein(10)

gd luck to all □

Bio 302

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Mujhey CRISPR protein waly 2 question aye or ak RNA processing or ak operon model or ak translation or ak mRNA per note ak splicing process or iss ki types or ak open complex wala?

Bio 302

Bio 302 8 am (final term)
Splicing pathway... CRISTR STANDS FOR??? termination in short... Rna processing .. rna has short life span so how we can analyzed through what process fr elongation short Operon modelbs yhi yad

Bio 302

302. 18/8/2017 CRISPRs
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Gene regulation. 10mrks
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Initiation complex..3mrks
RNA editing 10mrks.
Mcqs bhooot ajeeeb bs 2 3 mcqs past sy ay. R

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#subject: #bio302
12/08/2017
#final #term #paper Q----1----Crspr stands for?
Q----2-----Operon model?
Q-----3-----Gene regulation?
Q------4-----Splicing pathways?

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Q------5------Splisosome?
Q------6------Short life of mRNA? Q------7------Ctd
tail?

Bio 302

what is the role of regulatory 13/8 time 8:00 am
Rna processing mechanism
Rna editing
Gene expression Operon
model
Crispr stands for
Types of splicing explain its two elements ?

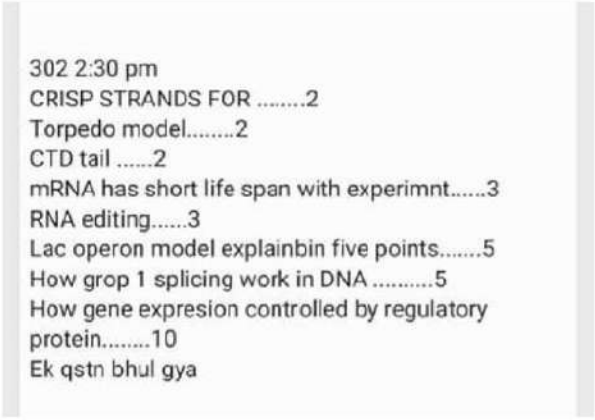
Bio 302
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Bio 302


302 2:30 pm
CRISP STRANDS FOR 2
Torpedo model.....2
CTD tail 2
mRNA has short life span with experimnt.....3
RNA editing.....3
Lac operon model explainbin five points.....5
How grop 1 splicing work in DNA 5
How gene expresion controlled by regulatory protein.....10
Ek qstn bhul gya

Bio 302

Bio 302,
08:00, 19 Aug 17

Describe Terminate sequence ? 2
CRISPRs stands for ? 2
Two protein in E.Coli that are necessary for lactose metabolism? 2 How many classes of splicing found in the cell. Enlist, 3
mRNA short life spam, Name of Experiment, description of Experiment? 3
RNA processing events? 5
Operon Model ? 5
CRISPRs stands for, its work, application & implication of them ? 10
Splicing pathway, discuss individually? 10

Bio 302

Bio 302 final term 08:00 , 20 aug
MCQs+ Subjective
36)any DNA molecule that has the ability to replicate in aappropriate host cell to which the desired gene are integrated for cloning is called as a)plasmid b)linker c) vactor d)adaptor
35) regulatory domains of most activator interact with the
a) transcription factor complex b)RNA polymerase c)repressor d)regulatory factor complex
34)which type of transcription termination requires formation of stem loop secondary structure in RNAs?
a) p dependent b)p independent c) p independent & p dependent d)polymerase independent
33)which is the main function of tRNA in the relation of protein synthesis

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a)inhibit protein synthesis b)proofreading c)identities amino acid d)identities amino acid and transport to the ribosome

32)in context with the mechanism of RNA editing one important clue about the mechanism of RNA editing is that partially edited transcript have been isolated and these are always edited at their

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31) eukaryotic cell always use ---------- as the start codon. a)5'AAG3' b) 5'AUC3' c) 5'AUG3' d) 5'TUG3'

30) a variable loop sits b/w anticodon loop and the Uloop and its name implies,

a) varies in size from 2 to 20 bases b) varies in size from 4 to 22 bases c) varies in size from 3 to 21 bases d) varies in size from 5 to 23 bases

29)the small subunit of ribosomes contains the -----content in which charged tRNA read or ---- the codon units of the mRNA

a) encoding, encode b) decoding, decode c)catalytic, catalysis d)production, producers

28) it is supposed that in ribosomes peptidly transferaseis entirely composed of a)DNA b)RNA c)proteins d)lipids

27)due to failure in proper nascent polypeptide process and folding the accomolation of toxic proteins which result from misfolding may lead to servant neurological disorder all the following axcept:

a) Alzhmir disease b)hungiston disease c)paekinson disease d) creutzfeldh Jacobs syndrome

26)CRISPRs consist repeat sequence ----- long .....(mazeed 2lines ke statement or b the)

a)each 10 bp b) each 20 bp c) each 30 bp d) each 40 bp 25)although m RNA short life time may seem wasteful it has

a) an important catalytic function b) an important regulatory function c) an important survival function d) an important revolutionary function

24) Monod and Jacob proposed the operon model in 1961.

23) in catabolite repression the system requires for proteins which following of the two are also component of other sugar transportersystem and therefor are unlikely to be direct participates in a glucose specific phenomenon ?

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a) 4 structural gene designated as trp D, trpC ,trpB, trpA, b) 5 structural gene designated as trpE, trp D, trpC ,trpB, trpA c) 6 structural gene designated as trpF, trpE, trp D, trpC ,trpB, trpA d) 7 structural gene designated as trpG, trpF, trpE, trp D, trpC ,trpB, trpA

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Subjective

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